

POWER BI ASSIGNMENT 1

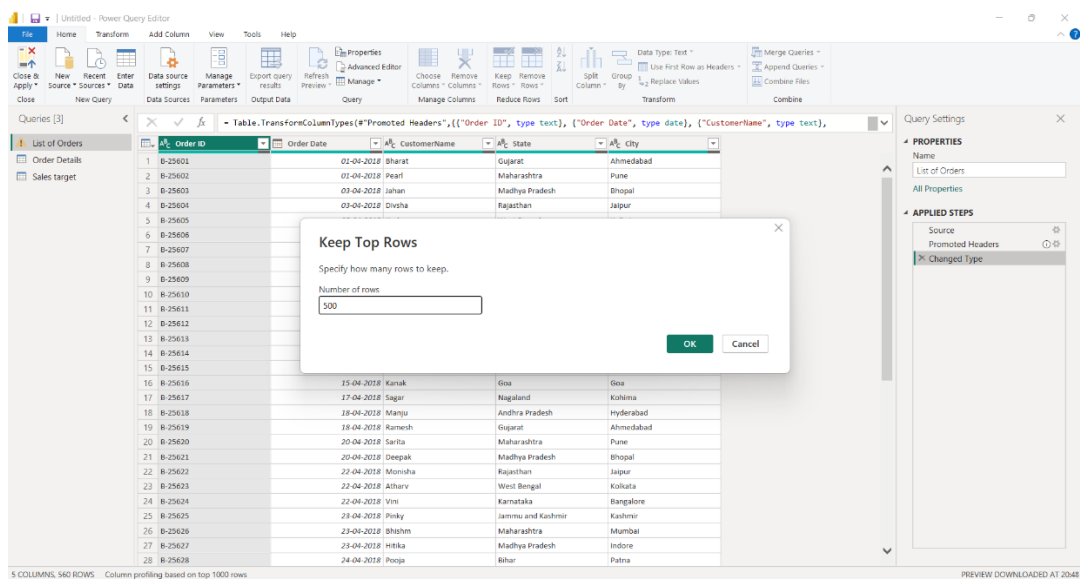
DATA TRANSFORMATION AND DATA MODELLING

I imported the required CSV files into Power BI and loaded the datasets. We then clicked **Transform Data** to open Power Query Editor for transforming the data.

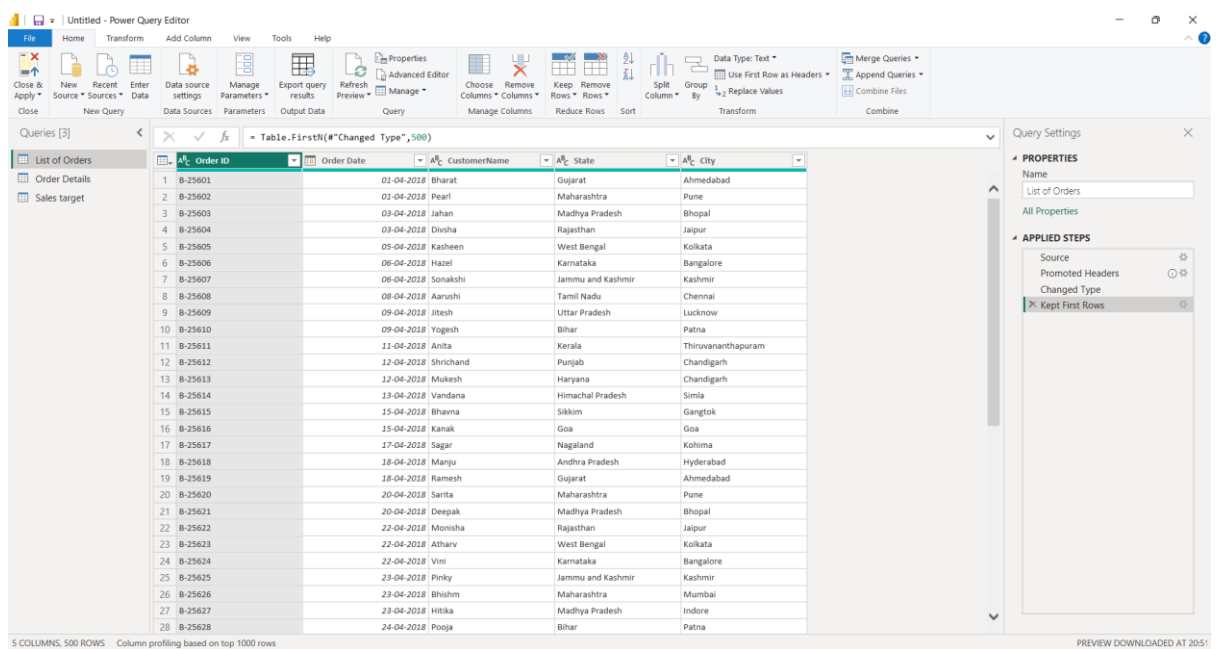
DATA TRANSFORMATION

- Restrict “List of Orders” to only the first 500 rows.

I opened **List of Orders** in Power Query, selected **Keep Rows** → **Keep Top Rows**, and entered **500** to keep only the first 500 rows.

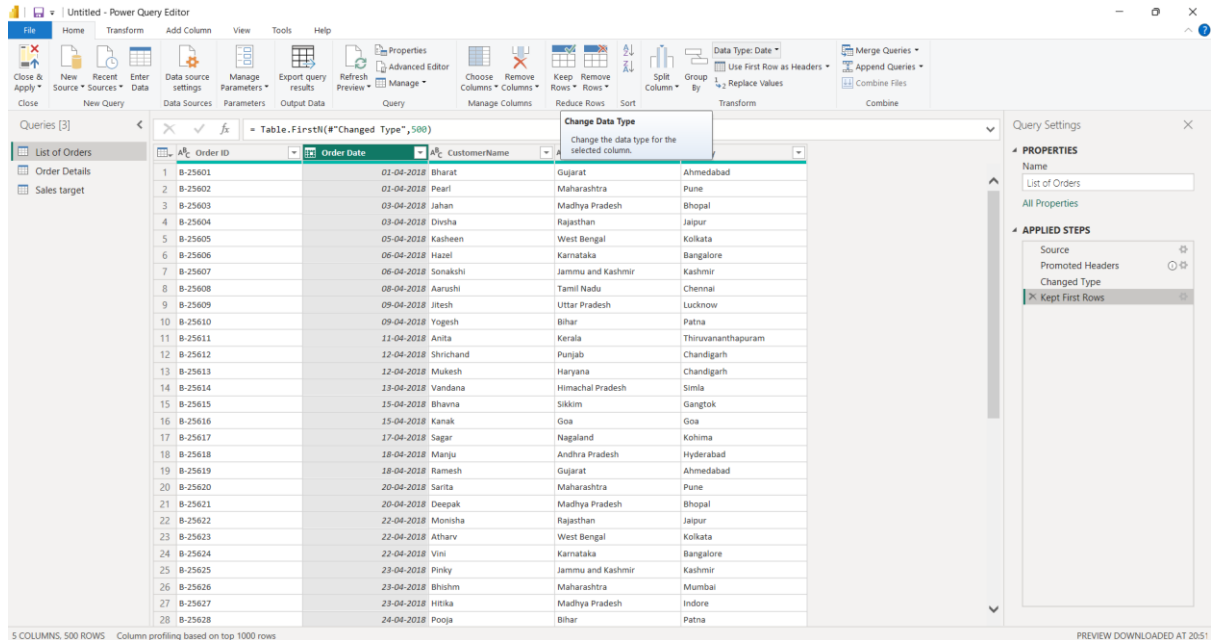


The result is as follows:



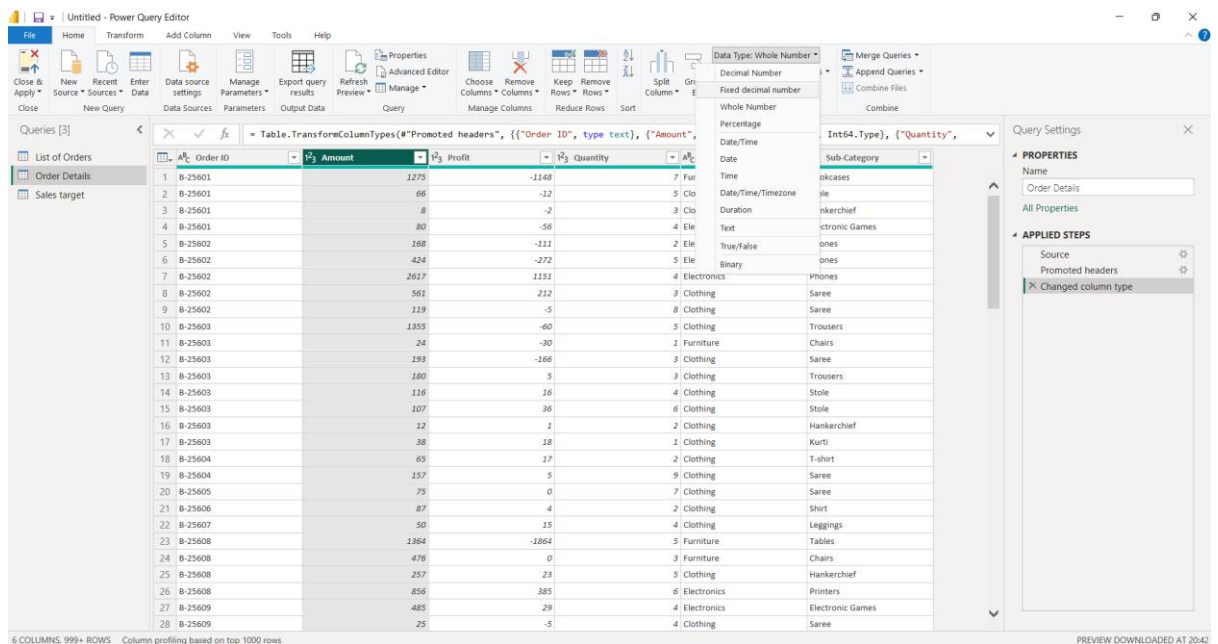
- Ensure the “Order Date” column in the “List of Orders” table is set to data type ‘Date’.

I selected the **Order Date** column and changed its data type to **Date** using the **Data Type** option in Power Query.



- Change the date type of “Amount” and “Target” columns to “Fixed Decimal Number”.

I selected the **Amount** and **Target** columns and changed their data type to **Fixed Decimal Number** using the **Data Type** option in Power Query.



3 COLUMNS, 36 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 2043

- Merge the “customer Name” column into proper case, ensuring consistent capitalisation for each word.

I selected the **Customer Name** column and changed the text format to **Capitalize Each Word (Proper Case)** to ensure consistent capitalization.

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Queries [3]

List of Orders

Order Details

Sales target

Table.TransformColumns(#"Kept First Rows",{"CustomerName", Text.Proper, type text})

Order ID	Order Date	CustomerName	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla
15	15-04-2018	Bhavina	Sikkim	Gangtok
16	15-04-2018	Kanak	Goa	Goa
17	17-04-2018	Sagar	Nagaland	Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata
24	22-04-2018	Yini	Karnataka	Bangalore
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	23-04-2018	Bhishm	Maharashtra	Mumbai
27	23-04-2018	Hitika	Madhya Pradesh	Indore
28	24-04-2018	Pooja	Bihar	Patna

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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- Merge the “State” and “City” columns into proper case, ensuring consistent capitalization for each word.

I selected the **State** and **City** columns and changed both to **Capitalize Each Word (Proper Case)** to ensure consistent capitalization as well.

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File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Queries [3]

List of Orders

Order Details

Sales target

Table.TransformColumns(#"Kept First Rows",{"CustomerName", Text.Proper, type text})

Merge Columns

Concatenate the currently selected columns into one column.

Order ID	Order Date	CustomerName	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla
15	15-04-2018	Bhavina	Sikkim	Gangtok
16	15-04-2018	Kanak	Goa	Goa
17	17-04-2018	Sagar	Nagaland	Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata
24	22-04-2018	Yini	Karnataka	Bangalore
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	23-04-2018	Bhishm	Maharashtra	Mumbai
27	23-04-2018	Hitika	Madhya Pradesh	Indore
28	24-04-2018	Pooja	Bihar	Patna

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 21:01

The screenshot shows the Power Query Editor interface. The main table has the following data:

Order ID	Order Date	Customer Name	Location
1	01-04-2018	Bharat	Ahmedabad, Gujarat
2	01-04-2018	Pearl	Pune, Maharashtra
3	03-04-2018	Jahan	Bhopal, Madhya Pradesh
4	03-04-2018	Divsha	Jaipur, Rajasthan
5	05-04-2018	Kasheen	Kolkata, West Bengal
6	06-04-2018	Hazel	Bangalore, Karnataka
7	06-04-2018	Sonakshi	Kashmir, Jammu and Kashmir
8	08-04-2018	Aarushi	Chennai, Tamil Nadu
9	09-04-2018	Jitesh	Lucknow, Uttar Pradesh
10	09-04-2018	Yogesh	Patna, Bihar
11	11-04-2018	Anita	Thiruvananthapuram, Kerala
12	12-04-2018	Shrichand	Chandigarh, Punjab
13	12-04-2018	Mukesh	Chandigarh, Haryana
14	13-04-2018	Vandana	Simla, Himachal Pradesh
15	15-04-2018	Bhavina	Gangtok, Sikkim
16	15-04-2018	Kanak	Goa, Goa
17	17-04-2018	Sagar	Kohima, Nagaland
18	18-04-2018	Manju	Hyderabad, Andhra Pradesh
19	18-04-2018	Ramesh	Ahmedabad, Gujarat
20	20-04-2018	Sarita	Pune, Maharashtra
21	20-04-2018	Deepak	Bhopal, Madhya Pradesh
22	22-04-2018	Monisha	Jaipur, Rajasthan
23	22-04-2018	Attharv	Kolkata, West Bengal
24	22-04-2018	Vini	Bangalore, Karnataka
25	23-04-2018	Pinky	Kashmir, Jammu and Kashmir
26	23-04-2018	Bhishm	Mumbai, Maharashtra
27	23-04-2018	Hitika	Indore, Madhya Pradesh
28	24-04-2018	Pooja	Patna, Bihar

- Create a new custom column names “Profit Margin” as the percentage of “Profit” divided by “Amount”.

I went to **Add Column** → **Custom Column**, named it **Profit Margin**, and used the formula **Profit ÷ Amount** to calculate the percentage.

The screenshot shows the Power Query Editor interface with the 'Custom Column' dialog box open. The dialog box has the following details:

- New column name:** Profit Margin
- Custom column formula:** = [Profit] / [Amount]
- Available columns:** Order ID, Amount, Profit, Quantity, Category, Sub-Category
- Confirmation:** No syntax errors have been detected.

The background table shows the following data:

Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	87.00	4	2	Clothing	Shirt
2	50.00	15	4	Clothing	Leggings
3	1,364.00	-1864	5	Furniture	Tables
4	476.00	0	3	Furniture	Chairs
5	257.00	23	5	Clothing	Hankerchief
6	856.00	385	6	Electronics	Printers
7	485.00	29	4	Electronics	Electronic Games
8	25.00	-5	4	Clothing	Saree

- Add a new conditional column named "Profit Status" based on the values in the "Profit" column. The conditions are as follows: if the profit is less than 0, the label should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the profit is greater than 0, the label should be "Profit".

I went to **Add Column** → **Conditional Column** in Power Query.

The screenshot shows the Power Query Editor interface. The main area displays a data table with the following columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, and Profit Margin. The 'Profit' column contains values ranging from -148 to 385. The 'Profit Margin' column contains values ranging from 0.12 to 1.2. The 'Category' column lists various product types like Furniture, Electronics, Clothing, etc. The 'Sub-Category' column lists specific products like Bookcases, Stole, Hankerchief, etc.

I entered **Profit Status** as the new column name and added the required conditions: **Profit < 0** → **Loss**, **Profit = 0** → **Break-Even**, and **Profit > 0** → **Profit** and clicked **OK** to create the **Profit Status** column.

The screenshot shows the 'Add Conditional Column' dialog box in the Power Query Editor. The dialog box has the following fields and options:

- New column name:** Profit Status
- Column Name:** Profit
- Operator:** is less than, equals, is greater than
- Value:** 0
- Output:** Loss, Break-Even, Profit

The 'Add Clause' button is visible at the bottom of the dialog box. The background shows the same data table as the previous screenshot.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples * Column Function Duplicate Column General

Conditional Column Index Column Merge Columns Extract Format Parse From Text From Number From Date & Time

Statistics Standard Scientific Trigonometry Rounding Information Date Time Duration

Queries [3]

List of Orders

Order Details

Sales target

Table.AddColumn(#"Changed Type of Profit Margin", "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then

	P3 Profit	P3 Quantity	A6 Category	A6 Sub-Category	L2 Profit Margin	P3 Profit Status
1474	6.00	1	1 Clothing	Kurti	0.16666667	Profit
1475	45.00	9	3 Clothing	Leggings	0.2	Profit
1476	103.00	46	2 Clothing	Saree	0.446601942	Profit
1477	140.00	56	4 Clothing	Shirt	0.4	Profit
1478	88.00	11	3 Electronics	Accessories	0.125	Profit
1479	451.00	25	3 Electronics	Phones	0.055432373	Profit
1480	264.00	-26	3 Clothing	Trousers	-0.098484848	Loss
1481	97.00	12	2 Clothing	Hankerschief	0.12371134	Profit
1482	14.00	5	1 Clothing	Hankerschief	0.357142857	Profit
1483	19.00	8	2 Clothing	Hankerschief	0.421052632	Profit
1484	39.00	18	2 Clothing	Leggings	0.461538462	Profit
1485	185.00	-26	6 Furniture	Chairs	-0.140540541	Loss
1486	663.00	-212	5 Electronics	Printers	-0.19758679	Loss
1487	671.00	-309	5 Electronics	Electronic Games	-0.460506706	Loss
1488	82.00	8	3 Electronics	Accessories	0.097560976	Profit
1489	497.00	179	3 Furniture	Chairs	0.360160996	Profit
1490	96.00	48	5 Clothing	Leggings	0.5	Profit
1491	409.00	86	3 Clothing	Saree	0.210268949	Profit
1492	59.00	15	2 Clothing	T-shirt	0.254237288	Profit
1493	46.00	14	5 Clothing	Skirt	0.304347826	Profit
1494	9.00	3	1 Clothing	Skirt	0.333333333	Profit
1495	207.00	37	4 Clothing	Hankerschief	0.178743961	Profit
1496	835.00	267	5 Electronics	Phones	0.319760479	Profit
1497	2,366.00	552	5 Clothing	Trousers	0.23305156	Profit
1498	828.00	230	2 Furniture	Chairs	0.277777778	Profit
1499	34.00	10	2 Clothing	T-shirt	0.294117647	Profit
1500	72.00	16	2 Clothing	Shirt	0.222222222	Profit

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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MERGE DATA (JOINS)

- Merge the "List of Orders" and "Order Details" tables into a new single table named "Orders Data" based on the "Order ID" relationship.

I selected **Merge Queries**, chose **List of Orders** and **Order Details**, selected **Order ID** as the matching column, chose the appropriate **Join Kind**, and clicked **OK**.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Manage Parameters Export query results Output Data

Properties Advanced Editor

Data Type: Text Use First Row as Headers Merge Queries Append Queries

Query Settings

Properties Name List of Orders

Applied Steps Source Promoted Headers Changed Type Kept First Rows Capitalized Each Word - Cust...

Merged Columns

Table.CombineC

Order ID

Order Details

Sales target

Table.CombineC

Order ID

Order Date

CustomerName

Location

Order ID

Amount

Profit

Quantity

Category

Sub-Category

Profit Margin

Profit Status

Join Kind

Left Outer (all from first, matching from second)

Use fuzzy matching to perform the merge

Fuzzy matching options

The selection matches 500 of 500 rows from the first table.

OK Cancel

23-04-2018 Bhishm Mumbai, Maharashtra

23-04-2018 Hitika Indore, Madhya Pradesh

24-04-2018 Pooja Patna, Bihar

4 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Data Close New Query

Data source settings Manage Parameters Export query results Refresh Preview * Properties Advanced Editor Manage * Query Choose Columns * Remove Columns * Manage Columns Reduce Rows Sort Split Column * Group By * Data Type: Text * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [4] List of Orders Order Details Sales target Orders Data

Table.NestedJoin(List of Orders, "Order ID", Order Details, "Order ID", JoinKind.LeftOuter)

Order ID	Order Date	CustomerName	Location	Order Details
1 B-25601	01-04-2018	Bharat	Ahmedabad, Gujarat	Table
2 B-25602	01-04-2018	Pearl	Pune, Maharashtra	Table
3 B-25603	03-04-2018	Jahan	Bhopal, Madhya Pradesh	Table
4 B-25604	03-04-2018	Divsha	Jaipur, Rajasthan	Table
5 B-25605	05-04-2018	Kasheen	Kolkata, West Bengal	Table
6 B-25606	06-04-2018	Hazel	Bangalore, Karnataka	Table
7 B-25607	06-04-2018	Sonakshi	Kashmir, Jammu and Kashmir	Table
8 B-25608	08-04-2018	Aarushi	Chennai, Tamil Nadu	Table
9 B-25609	09-04-2018	Jitesh	Lucknow, Uttar Pradesh	Table
10 B-25610	09-04-2018	Yogesh	Patna, Bihar	Table
11 B-25611	11-04-2018	Anita	Thiruvananthapuram, Kerala	Table
12 B-25612	12-04-2018	Shrichand	Chandigarh, Punjab	Table
13 B-25613	12-04-2018	Mukesh	Chandigarh, Haryana	Table
14 B-25614	13-04-2018	Vandana	Simla, Himachal Pradesh	Table
15 B-25615	15-04-2018	Bhavina	Gangtok, Sikkim	Table
16 B-25616	15-04-2018	Kanak	Goa, Goa	Table
17 B-25617	17-04-2018	Sagar	Kohima, Nagaland	Table
18 B-25618	18-04-2018	Manju	Hyderabad, Andhra Pradesh	Table
19 B-25619	18-04-2018	Ramesh	Ahmedabad, Gujarat	Table
20 B-25620	20-04-2018	Sarita	Pune, Maharashtra	Table
21 B-25621	20-04-2018	Deepak	Bhopal, Madhya Pradesh	Table
22 B-25622	22-04-2018	Monisha	Jaipur, Rajasthan	Table
23 B-25623	22-04-2018	Atharv	Kolkata, West Bengal	Table
24 B-25624	22-04-2018	Vini	Bangalore, Karnataka	Table
25 B-25625	23-04-2018	Pinky	Kashmir, Jammu and Kashmir	Table
26 B-25626	23-04-2018	Bhishm	Mumbai, Maharashtra	Table
27 B-25627	23-04-2018	Hritika	Indore, Madhya Pradesh	Table
28 B-25628	24-04-2018	Pooja	Patna, Bihar	Table

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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I expanded the merged column to bring the required details into one table and named the new table **Orders Data**.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Data Close New Query

Data source settings Manage Parameters Export query results Refresh Preview * Properties Advanced Editor Manage * Query Choose Columns * Remove Columns * Manage Columns Reduce Rows Sort Split Column * Group By * Data Type: Table * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [4] List of Orders Order Details Sales target Orders Data

Table.NestedJoin(List of Orders, "Order ID", Order Details, "Order ID", JoinKind.LeftOuter)

Search Columns to Expand

Expand Aggregate

(Select All Columns)

Order ID

Amount

Profit

Quantity

Sub-Category

Profit Margin

Profit Status

Use original column name as prefix

OK Cancel

Order ID	Order Date	CustomerName	Location	Order Details
1 B-25601	01-04-2018	Bharat	Ahmedabad, Gujarat	Table
2 B-25602	01-04-2018	Pearl	Pune, Maharashtra	Table
3 B-25603	03-04-2018	Jahan	Bhopal, Madhya Pradesh	Table
4 B-25604	03-04-2018	Divsha	Jaipur, Rajasthan	Table
5 B-25605	05-04-2018	Kasheen	Kolkata, West Bengal	Table
6 B-25606	06-04-2018	Hazel	Bangalore, Karnataka	Table
7 B-25607	06-04-2018	Sonakshi	Kashmir, Jammu and Kashmir	Table
8 B-25608	08-04-2018	Aarushi	Chennai, Tamil Nadu	Table
9 B-25609	09-04-2018	Jitesh	Lucknow, Uttar Pradesh	Table
10 B-25610	09-04-2018	Yogesh	Patna, Bihar	Table
11 B-25611	11-04-2018	Anita	Thiruvananthapuram, Kerala	Table
12 B-25612	12-04-2018	Shrichand	Chandigarh, Punjab	Table
13 B-25613	12-04-2018	Mukesh	Chandigarh, Haryana	Table
14 B-25614	13-04-2018	Vandana	Simla, Himachal Pradesh	Table
15 B-25615	15-04-2018	Bhavina	Gangtok, Sikkim	Table
16 B-25616	15-04-2018	Kanak	Goa, Goa	Table
17 B-25617	17-04-2018	Sagar	Kohima, Nagaland	Table
18 B-25618	18-04-2018	Manju	Hyderabad, Andhra Pradesh	Table
19 B-25619	18-04-2018	Ramesh	Ahmedabad, Gujarat	Table
20 B-25620	20-04-2018	Sarita	Pune, Maharashtra	Table
21 B-25621	20-04-2018	Deepak	Bhopal, Madhya Pradesh	Table
22 B-25622	22-04-2018	Monisha	Jaipur, Rajasthan	Table
23 B-25623	22-04-2018	Atharv	Kolkata, West Bengal	Table
24 B-25624	22-04-2018	Vini	Bangalore, Karnataka	Table
25 B-25625	23-04-2018	Pinky	Kashmir, Jammu and Kashmir	Table
26 B-25626	23-04-2018	Bhishm	Mumbai, Maharashtra	Table
27 B-25627	23-04-2018	Hritika	Indore, Madhya Pradesh	Table
28 B-25628	24-04-2018	Pooja	Patna, Bihar	Table

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 21:32

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns * Columns * Manage Columns Reduce Rows Sort Split Column * Group By Data Type: Any * Use First Row as Headers * Replace Values * Merge Queries * Append Queries * Combine Files * Combine

Queries [4] List of Orders Order Details Sales target Orders Data

Table.ExpandTableColumn(Source, "Order Details", {"Amount", "Profit", "Quantity", "Category", "Sub-Category", "Profit Margin", "Profit"})

Order ID	Order Date	CustomerName	Location	Amount	Profit	Quantity
1	01-04-2018	Bharat	Ahmedabad, Gujarat	1,275.00	-1148	
2	01-04-2018	Bharat	Ahmedabad, Gujarat	66.00	-12	
3	01-04-2018	Bharat	Ahmedabad, Gujarat	8.00	-2	
4	01-04-2018	Bharat	Ahmedabad, Gujarat	80.00	-56	
5	01-04-2018	Pearl	Pune, Maharashtra	168.00	-111	
6	01-04-2018	Pearl	Pune, Maharashtra	424.00	-272	
7	01-04-2018	Pearl	Pune, Maharashtra	2,617.00	1151	
8	01-04-2018	Pearl	Pune, Maharashtra	561.00	212	
9	01-04-2018	Pearl	Pune, Maharashtra	119.00	-5	
10	03-04-2018	Jahan	Bhopal, Madhya Pradesh	1,355.00	-60	
11	03-04-2018	Jahan	Bhopal, Madhya Pradesh	24.00	-30	
12	03-04-2018	Jahan	Bhopal, Madhya Pradesh	193.00	-166	
13	03-04-2018	Jahan	Bhopal, Madhya Pradesh	180.00	5	
14	03-04-2018	Jahan	Bhopal, Madhya Pradesh	116.00	16	
15	03-04-2018	Jahan	Bhopal, Madhya Pradesh	107.00	36	
16	03-04-2018	Jahan	Bhopal, Madhya Pradesh	12.00	1	
17	03-04-2018	Jahan	Bhopal, Madhya Pradesh	38.00	18	
18	03-04-2018	Divdha	Jaipur, Rajasthan	65.00	17	
19	03-04-2018	Divdha	Jaipur, Rajasthan	157.00	5	
20	05-04-2018	Kasheen	Kolkata, West Bengal	75.00	0	
21	06-04-2018	Hazel	Bangalore, Karnataka	87.00	4	
22	06-04-2018	Sonakshi	Kashmir, Jammu and Kashmir	50.00	15	
23	08-04-2018	Aarushi	Chennai, Tamil Nadu	1,364.00	-1864	
24	08-04-2018	Aarushi	Chennai, Tamil Nadu	476.00	0	
25	08-04-2018	Aarushi	Chennai, Tamil Nadu	257.00	23	
26	08-04-2018	Aarushi	Chennai, Tamil Nadu	856.00	385	
27	08-04-2018	Aarushi	Chennai, Tamil Nadu	856.00	385	
28	09-04-2018	Jitesh	Lucknow, Uttar Pradesh	485.00	29	

11 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 21:44

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns * Columns * Manage Columns Reduce Rows Sort Split Column * Group By Data Type: Any * Use First Row as Headers * Replace Values * Merge Queries * Append Queries * Combine Files * Combine

Queries [4] List of Orders Order Details Sales target Orders Data

Table.ExpandTableColumn(Source, "Order Details", {"Amount", "Profit", "Quantity", "Category", "Sub-Category", "Profit Margin", "Profit"})

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
1	1,275.00	-1148		7 Furniture	Bookcases	-0.900392157	Loss
2	66.00	-12		5 Clothing	Stole	-0.181818182	Loss
3	8.00	-2		3 Clothing	Hankerchief	-0.25	Loss
4	80.00	-56		4 Electronics	Electronic Games	-0.7	Loss
5	168.00	-111		2 Electronics	Phones	-0.660714286	Loss
6	424.00	-272		5 Electronics	Phones	-0.641509434	Loss
7	2,617.00	1151		4 Electronics	Phones	0.439816584	Profit
8	561.00	212		3 Clothing	Saree	0.377896613	Profit
9	119.00	-5		8 Clothing	Saree	-0.042016807	Loss
10	1,355.00	-60		5 Clothing	Trousers	-0.044280443	Loss
11	24.00	-30		1 Furniture	Chairs	-1.25	Loss
12	193.00	-166		3 Clothing	Saree	-0.860103627	Loss
13	180.00	5		3 Clothing	Trousers	0.027777778	Profit
14	116.00	16		4 Clothing	Stole	0.137931084	Profit
15	107.00	36		6 Clothing	Stole	0.336448598	Profit
16	12.00	1		2 Clothing	Hankerchief	0.083333333	Profit
17	38.00	18		1 Clothing	Kurti	0.473684211	Profit
18	65.00	17		2 Clothing	T-shirt	0.261538462	Profit
19	157.00	5		9 Clothing	Saree	0.031847134	Profit
20	75.00	0		7 Clothing	Saree	0	Break-Even
21	87.00	4		2 Clothing	Shirt	0.045977021	Profit
22	50.00	15		4 Clothing	Leggings	0.3	Profit
23	1,364.00	-1864		5 Furniture	Tables	-1.366568915	Loss
24	476.00	0		3 Furniture	Chairs	0	Break-Even
25	257.00	23		5 Clothing	Hankerchief	0.089494163	Profit
26	856.00	385		6 Electronics	Printers	0.449766355	Profit
27	485.00	29		4 Electronics	Electronic Games	0.059793814	Profit

11 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 21:44

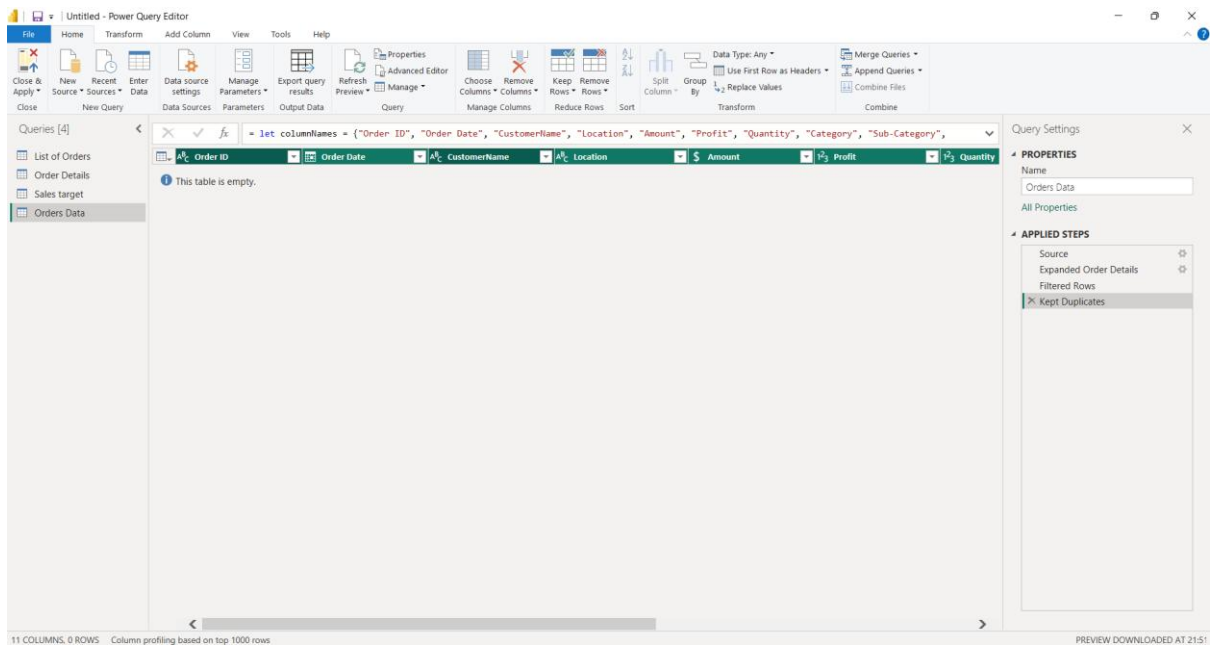
HANDLING MISSING DATA & DUPLICATE DATA:

- Identify missing values in the data and determine a strategy to address them.

I checked the **filters for each column** in Power Query and found that there were **no missing or blank values** in the data.

- Check for duplicate rows and define a strategy to handle duplicates.

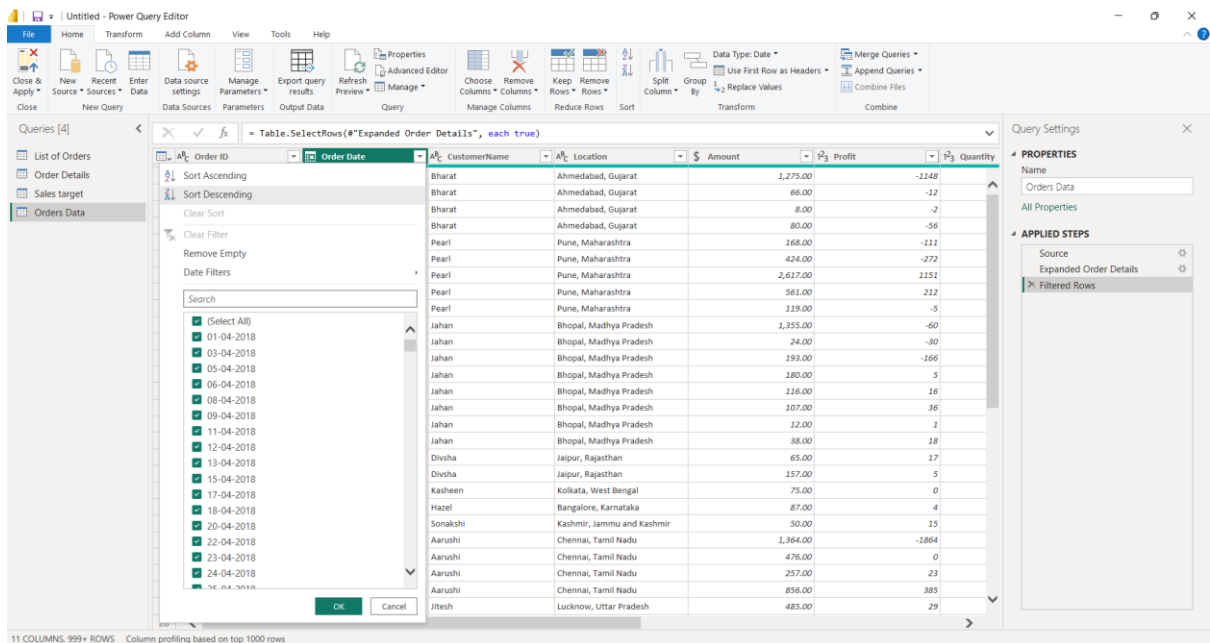
I checked for duplicates using **Remove Rows → Remove Duplicates** in Power Query and found that there were **no duplicate rows**, so no duplicates were removed.



SORTING & FILTERING DATA

- In the 'Orders Data' table, utilize sorting and filtering techniques on columns like Order Date, State or Category to analyze data based on specific criteria:
 - Sort the orders by Order Date in descending order to analyze recent trends.

I selected the **Order Date** column in the **Orders Data** table and used the filter dropdown to **Sort Descending**, arranging the orders from the most recent date to the oldest.



I selected the **Location** column, used **Extract** → **Text Before Delimiter**, and selected a comma (,) as the **delimiter** to extract the State.

The screenshot shows the Power Query Editor interface. The 'Queries' pane on the left lists 'List of Orders', 'Order Details', 'Sales target', and 'Orders Data'. The main area displays a table with columns: Order ID, Order Date, Location, Amount, Profit, and Quantity. The 'Location' column is selected, and the 'Extract' menu is open, showing 'Text Before Delimiter' as the chosen option. A tooltip indicates: 'Return the text that occurs after a delimiter.' The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list: 'Source', 'Expanded Order Details', 'Filtered Rows', and 'Sorted Rows order date'.

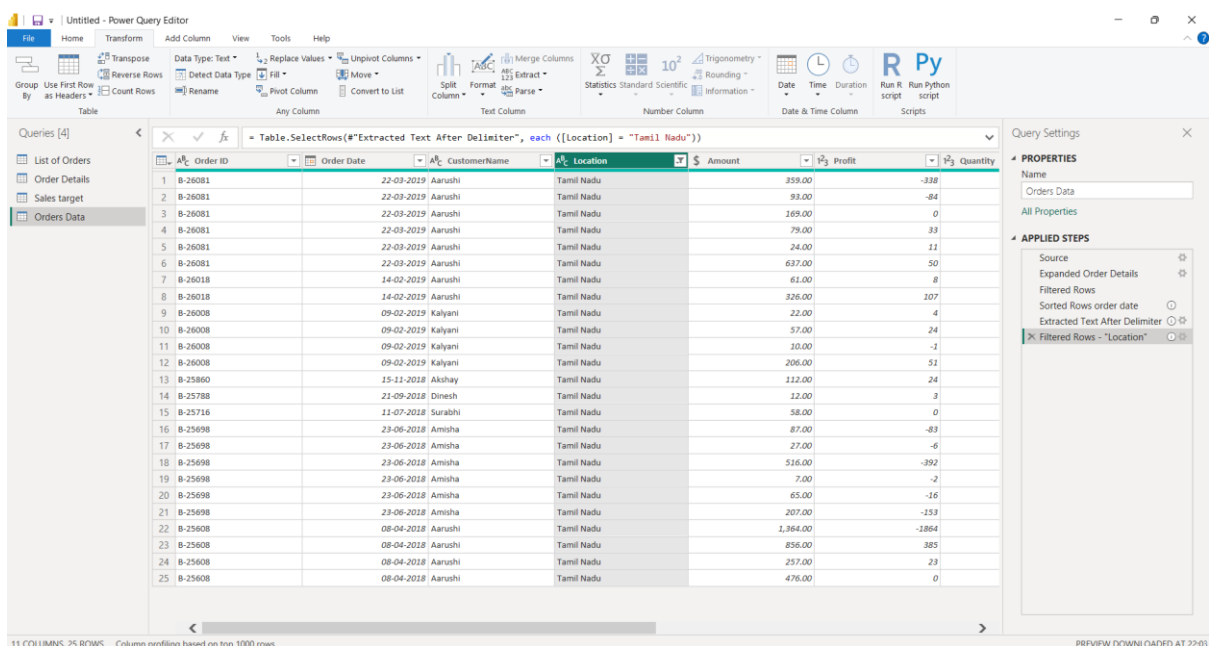
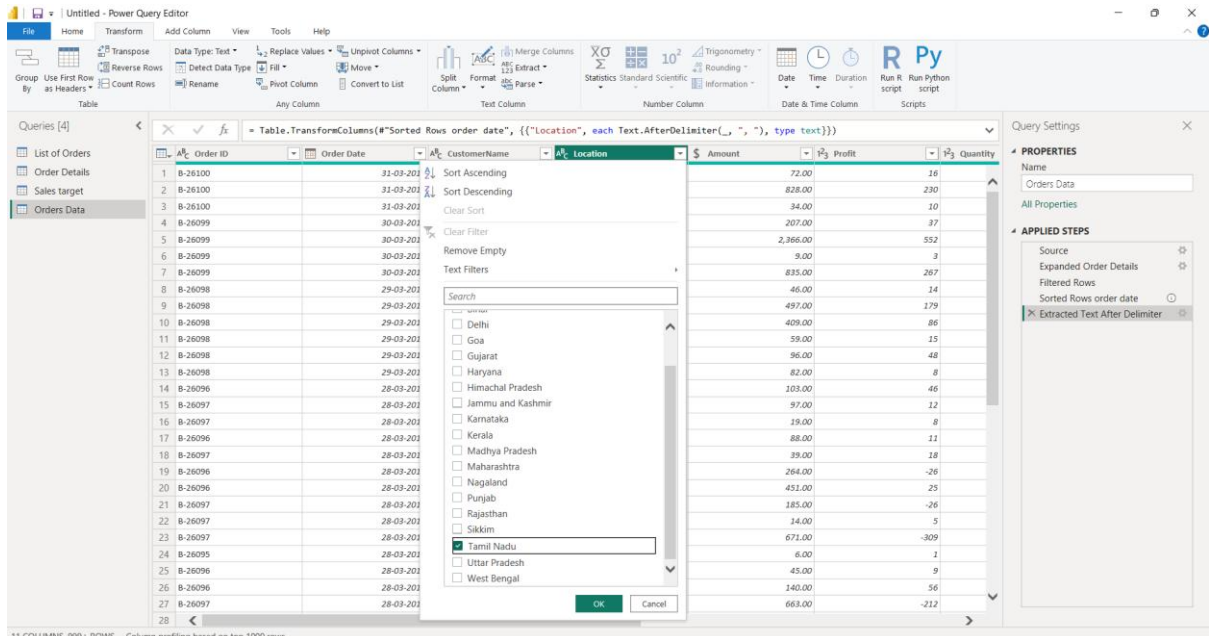
Order ID	Order Date	Location	Amount	Profit	Quantity
1	31-03-2019	Indore, Madhya Pradesh	72.00	16	16
2	31-03-2019	Indore, Madhya Pradesh	828.00	230	230
3	31-03-2019	Indore, Madhya Pradesh	34.00	10	10
4	30-03-2019	Mumbai, Maharashtra	207.00	37	37
5	30-03-2019	Mumbai, Maharashtra	2,866.00	552	552
6	30-03-2019	Mumbai, Maharashtra	9.00	3	3
7	30-03-2019	Mumbai, Maharashtra	835.00	267	267
8	29-03-2019	Kashmir, Jammu and Kashmir	46.00	14	14
9	29-03-2019	Kashmir, Jammu and Kashmir	497.00	179	179
10	29-03-2019	Kashmir, Jammu and Kashmir	409.00	86	86
11	29-03-2019	Kashmir, Jammu and Kashmir	59.00	15	15
12	29-03-2019	Kashmir, Jammu and Kashmir	96.00	48	48
13	29-03-2019	Kashmir, Jammu and Kashmir	82.00	8	8
14	28-03-2019	Kolkata, West Bengal	103.00	46	46
15	28-03-2019	Bangalore, Karnataka	97.00	12	12
16	28-03-2019	Bangalore, Karnataka	19.00	8	8
17	28-03-2019	Kolkata, West Bengal	88.00	11	11
18	28-03-2019	Bangalore, Karnataka	39.00	18	18
19	28-03-2019	Kolkata, West Bengal	264.00	-26	-26
20	28-03-2019	Kolkata, West Bengal	451.00	25	25
21	28-03-2019	Bangalore, Karnataka	185.00	-26	-26
22	28-03-2019	Bangalore, Karnataka	14.00	5	5
23	28-03-2019	Bangalore, Karnataka	671.00	-309	-309
24	28-03-2019	Jaipur, Rajasthan	6.00	1	1
25	28-03-2019	Kolkata, West Bengal	45.00	9	9
26	28-03-2019	Kolkata, West Bengal	140.00	56	56
27	28-03-2019	Bangalore, Karnataka	663.00	-212	-212

The screenshot shows the Power Query Editor interface. The 'Queries' pane on the left lists 'List of Orders', 'Order Details', 'Sales target', and 'Orders Data'. The main area displays a table with columns: Order ID, Order Date, CustomerName, Location, Amount, Profit, and Quantity. The 'Location' column is selected, and the 'Extract' menu is open, showing 'Text After Delimiter' as the chosen option. A dialog box titled 'Text After Delimiter' is open, prompting the user to 'Enter the delimiter that marks the beginning of what you would like to extract.' The 'Delimiter' field contains a comma (,). The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list: 'Source', 'Expanded Order Details', 'Filtered Rows', and 'Sorted Rows order date'.

Order ID	Order Date	CustomerName	Location	Amount	Profit	Quantity
1	31-03-2019	Hitika	Indore, Madhya Pradesh	72.00	16	16
2	31-03-2019	Hitika	Indore, Madhya Pradesh	828.00	230	230
3	31-03-2019	Hitika	Indore, Madhya Pradesh	34.00	10	10
4	30-03-2019	Bhishm	Mumbai, Maharashtra	207.00	37	37
5	30-03-2019	Bhishm	Mumbai, Maharashtra	2,866.00	552	552
6	30-03-2019	Bhishm	Mumbai, Maharashtra	9.00	3	3
7	30-03-2019	Bhishm	Mumbai, Maharashtra	835.00	267	267
8	29-03-2019	Pinky	Kashmir, Jammu and Kashmir	46.00	14	14
9	29-03-2019	Pinky	Kashmir, Jammu and Kashmir	497.00	179	179
10	29-03-2019	Pinky	Kashmir, Jammu and Kashmir	409.00	86	86
11	29-03-2019	Pinky	Kashmir, Jammu and Kashmir	59.00	15	15
12	29-03-2019	Pinky	Kashmir, Jammu and Kashmir	96.00	48	48
13	29-03-2019	Pinky	Kashmir, Jammu and Kashmir	82.00	8	8
14	28-03-2019	Atharv	Kolkata, West Bengal	103.00	46	46
15	28-03-2019	Vini	Bangalore, Karnataka	97.00	12	12
16	28-03-2019	Vini	Bangalore, Karnataka	19.00	8	8
17	28-03-2019	Atharv	Kolkata, West Bengal	88.00	11	11
18	28-03-2019	Vini	Bangalore, Karnataka	39.00	18	18
19	28-03-2019	Atharv	Kolkata, West Bengal	264.00	-26	-26
20	28-03-2019	Atharv	Kolkata, West Bengal	451.00	25	25
21	28-03-2019	Vini	Bangalore, Karnataka	185.00	-26	-26
22	28-03-2019	Vini	Bangalore, Karnataka	14.00	5	5
23	28-03-2019	Vini	Bangalore, Karnataka	671.00	-309	-309
24	28-03-2019	Monisha	Jaipur, Rajasthan	6.00	1	1
25	28-03-2019	Atharv	Kolkata, West Bengal	45.00	9	9
26	28-03-2019	Atharv	Kolkata, West Bengal	140.00	56	56
27	28-03-2019	Vini	Bangalore, Karnataka	663.00	-212	-212

- Filter the orders to focus only on a specific state (e.g., Tamil Nadu) for regional analysis.

I opened the **State** column filter and selected **Tamil Nadu** to display only orders from that state for regional analysis.



GROUPING & AGGREGATING DATA

- Duplicate the “Order Details” table and calculate the count of each Order ID, average profit by Category or total amount by Sub-Category.

I duplicated the **Order Details** table and used **Group By** to calculate the **average Profit for each Category**.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Queries [4]

Table.AddColumn(#"Changed Type of Profit Margin", "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then

Order Details

Order Details - Grouped

Properties

Applied Steps

Source

Promoted headers

Changed column type

Changed Type 'Amount' to FL

Added Custom Column

Changed Type of Profit Margin

Added Conditional Column n.0

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 21:25

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Properties

Choose Remove Columns Keep Remove Rows Reduce Rows Sort

Data Type: Whole Number Merge Queries Use First Row as Headers Replace Values Append Queries Combine Files Combine

Queries [5]

Table.AddColumn(#"Changed Type of Profit Margin", "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then

Order Details

Order Details - Grouped

Properties

Applied Steps

Source

Promoted headers

Changed column type

Changed Type 'Amount' to FL

Added Custom Column

Changed Type of Profit Margin

Added Conditional Column n.0

Group By

Specify the column to group by and the desired output.

Basic Advanced

Category

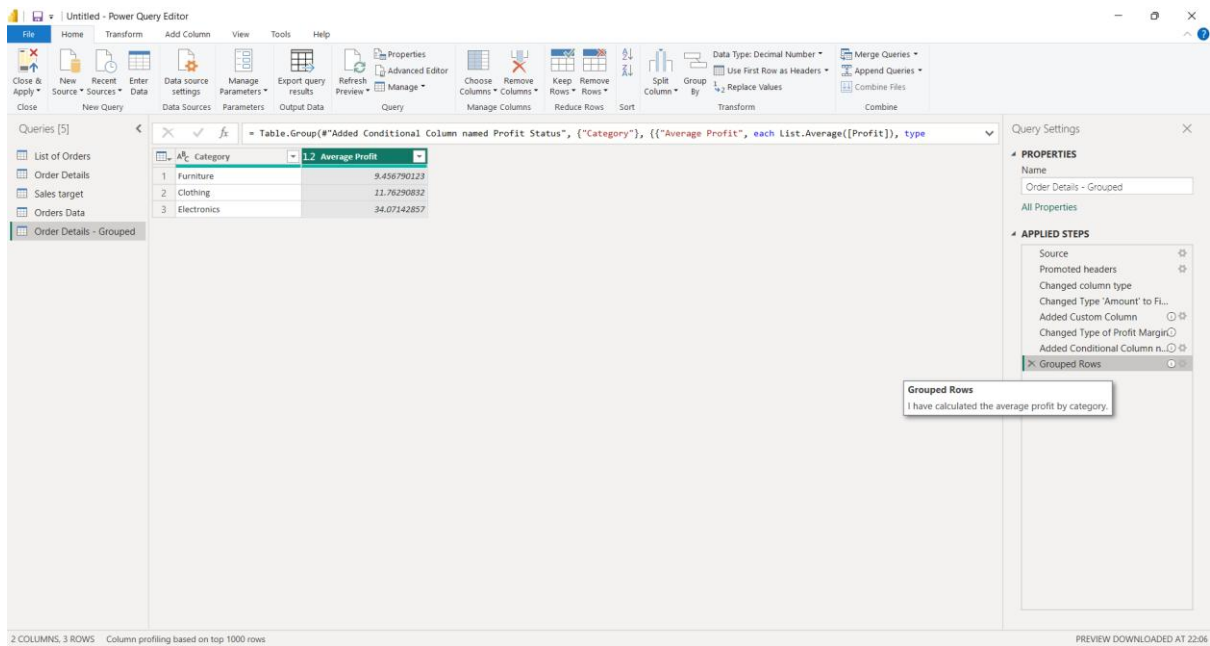
New column name Operation Column

Average Profit Average Profit

OK Cancel

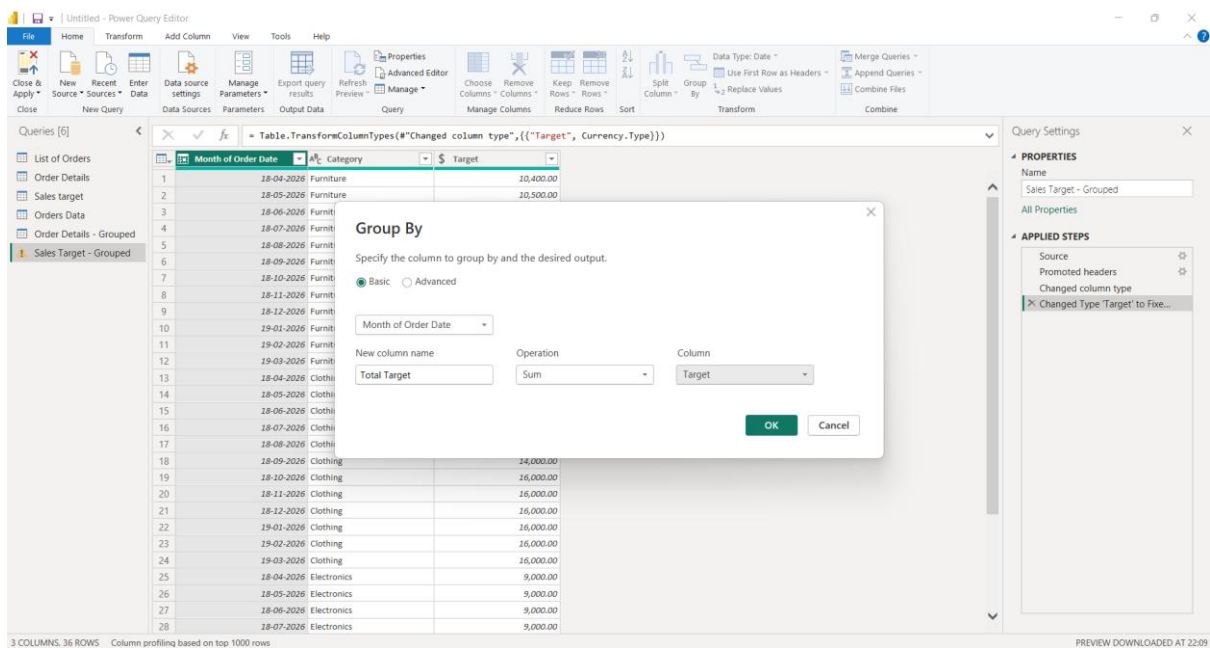
8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 22:04



- Duplicate the “Sales Target” table and aggregate the total target amount by Month of Order Date.

I duplicated the **Sales Target** table and used **Group By** to calculate the **total Target Amount** for each Month of Order Date.



I transformed the Order Date column to display only the Month of Order Date.

The screenshot displays the Power Query Editor interface. The main area shows a table with 12 columns and 12 rows. The first column is 'Month of Order Date', the second is 'Total Target', and the third is 'Total Target'. The data is grouped by 'Month of Order Date'. The 'Total Target' column is calculated as the sum of the 'Total Target' column for each month.

The 'Month of Order Date' column is highlighted, and a context menu is open, showing options like 'Date Only', 'Parse', 'Year', 'Month', 'Quarter', 'Week', 'Day', 'Combine Date and Time', 'Earliest', and 'Latest'. The 'Month' option is selected.

The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Source', 'Promoted headers', 'Changed column type', 'Changed Type 'Target' to Fix...', and 'Grouped Rows'.

The 'Properties' pane on the right shows the 'Name' field set to 'Sales Target - Grouped'.

The 'Table' view shows the following data:

Month of Order Date	Total Target	Total Target
18-04-2026	31400	31400
18-05-2026	31500	31500
18-06-2026	31600	31600
18-07-2026	31800	31800
18-08-2026	33900	33900
18-09-2026	34000	34000
18-10-2026	36100	36100
18-11-2026	36300	36300
18-12-2026	36400	36400
19-01-2026	43500	43500
19-02-2026	43600	43600
19-03-2026	43800	43800

The status bar at the bottom indicates '2 COLUMNS, 12 ROWS' and 'Column profiling based on top 1000 rows'.

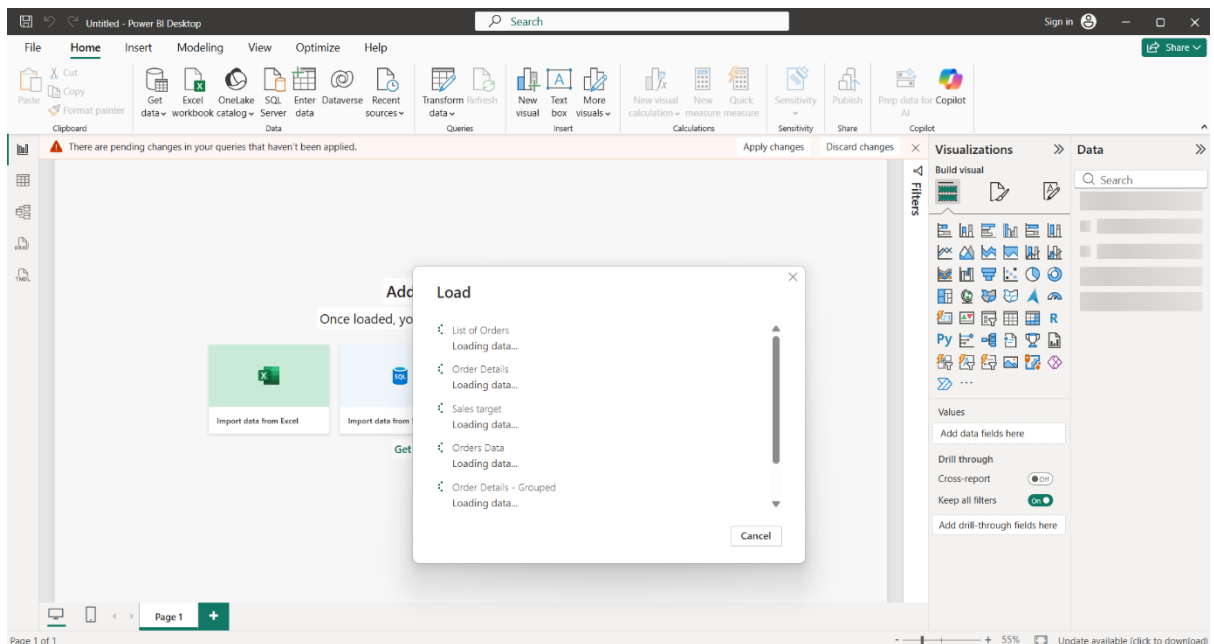
Power Query Editor interface showing a table with 12 rows and 2 columns. The first column is 'Month of Order Date' and the second is 'Total Target'. The data shows months from April to March with corresponding target values. A tooltip for 'Extracted Month Name' is visible, stating 'Extracted the name of the month from Date & Time column option'.

Month of Order Date	Total Target
1 April	31400
2 May	31500
3 June	31600
4 July	33800
5 August	33900
6 September	34000
7 October	36100
8 November	36300
9 December	36400
10 January	43500
11 February	43600
12 March	43800

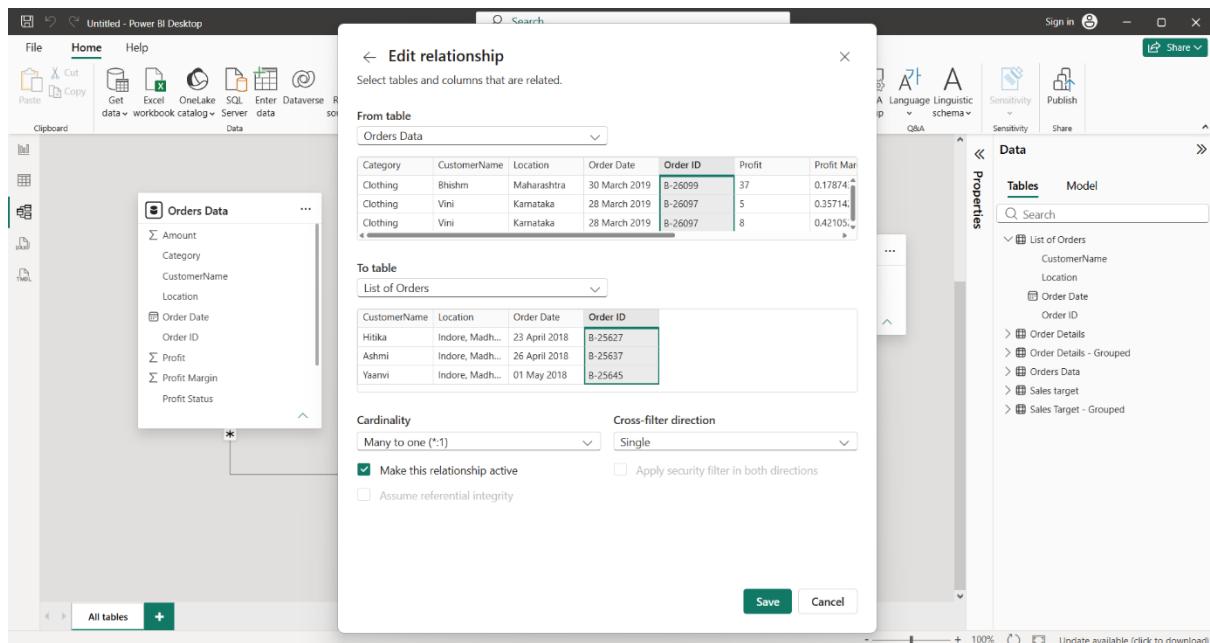
DATA MODELLING

- Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.

I applied all the required transformations in Power Query and then navigated to the **Model view** to establish relationships between the tables.



As soon as I loaded the data, Power BI **automatically established a relationship** between the Orders Data and List of Orders tables using the Order ID column.



- Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click "Manage relationships" and ensure this relationship is active.

I clicked **Manage Relationships** and selected **New Relationship** to create a relationship between Order Details and Sales Target using Category column.

